Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: *MILU* | ***Context:*** *Mymensingh Environmental Scientist — Cross-Regional South Asian Agricultural Knowledge*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU evaluates via accuracy on MCQ tasks using log-likelihood scoring for non-API models [Q48–Q51] and structured-JSON generative scoring for API models [Q53]. Both deployment and benchmark are text-based, satisfying the basic modality match. The MCQ output format does not require speech, vision, or other modalities that would create infrastructural barriers in Mymensingh (high text-only connectivity [WEB-20]). Two limitations reduce the score: (a) the authors acknowledge log-likelihood may yield different results than generation-based or CoT methods [Q85], which matters for the deployment's reasoning-heavy queries; and (b) MCQ output cannot capture the multi-step open-ended agricultural reasoning the deployment scenario requires (CRITICAL Concern 3, MAJOR Concern 7). For a LOWER-priority dimension where text modality is the main requirement, MILU is functionally adequate but suppresses richer output forms relevant to professional knowledge work.

ID	Evidence
Q48	“For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations.” (p.5)
Q49	“We use the log-likelihood method, where the probability of a given output string is computed by conditioning it on some provided input (Brown et al., 2020).” (p.5)
Q50	“"Specifically, the log-likelihood of an answer (a) given the question (x), i.e., logP(a\” (p.5)
Q51	“"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., argmax(logP(a1\” (p.5)
Q53	“We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing.” (p.5)
Q85	“Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting.” (p.9)
WEB-20	Bangladesh internet/mobile coverage adequate for text modality (https://datareportal.com/reports/digital-2024-bangladesh)

Strengths

- Text-based output modality is fully aligned with the deployment's text-only LLM access pattern [Q48, Q53]
- Reproducible evaluation via LM-EVALUATION-HARNESS [Q48] and structured JSON parsing for API models [Q53] enables consistent benchmarking across the frontier and regional Indic LLMs the user deploys
- 0/1/5-shot evaluation across base and instruct variants [Q45, Q46] provides multiple data points on model behavior, including the documented warning that instruct models may degrade with few-shot examples [Q62, Q63] — useful diagnostic signal for the deployment
- Consistent MCQ structure supports cross-language and cross-model comparability (DATASET Strength 4)

ID	Evidence
Q45	“Both the base versions and instruction fine-tuned variants of these models, wherever applicable, are evaluated to measure the improvements gained from fine-tuning.” (p.5)
Q46	“All models, except for proprietary models and LLAMA-3.1-405B, are tested under 0-shot, 1-shot, and 5-shot setups.” (p.5)
Q48	“For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations.” (p.5)
Q53	“We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing.” (p.5)

Q62	"In contrast, Instruct models exhibit more varied behavior, where models either stagnate or even degrade in performance." (p.6)
Q63	"This also aligns with expectations, as Instruct models are specifically fine-tuned as conversation assistants and may not respond well to the few-shot in-context examples format." (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (text MCQ answer-key selection) matches the deployment's text-based LLM access [Q48, Q53]. Modality match is good; format-richness match is partial — deployment expects open-ended reasoning, MCQ provides only label selection.

OF-2: Text-to-speech is not required by the deployment (text-based access only per YAML llm_access_modality). Not applicable.

OF-3: Target population is professional environmental scientists; literacy and accessibility are not constraints. Bangladesh national connectivity supports text modality [WEB-20, WEB-21]. Mymensingh hosts BAU, suggesting above-average professional digital infrastructure [WEB-7].

OF-4: Form mismatch is mild: log-likelihood vs. generative evaluation differences acknowledged by authors [Q85] mean MILU scores may not predict generation-mode deployment performance. MCQ format suppresses open-ended agricultural reasoning the deployment requires (MAJOR Concern 7). External validity for generation-mode deployment is partial.

ID	Evidence
Q48	"For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations." (p.5)
Q53	"We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing." (p.5)
Q85	"Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting." (p.9)
WEB-7	BAU as professional institutional context — text infrastructure supported (https://bau.edu.bd/)
WEB-20	Bangladesh internet/mobile coverage adequate for text modality (https://datareportal.com/reports/digital-2024-bangladesh)
WEB-21	Mobile-dominant connectivity in Bangladesh (https://en.wikipedia.org/wiki/Internet_in_Bangladesh)

Information Gaps

- Whether deployment workflow requires explicit chain-of-thought or generation-mode evaluation that would diverge from MILU's log-likelihood scoring [Q85]

Remediation

Gap 1: Log-likelihood MCQ scoring suppresses open-ended agricultural reasoning relevant to the deployment use case [Q85]

Recommendation: Add a generation-mode evaluation track with rubric-based scoring for a sub-sample of agriculturally relevant items, and include CoT prompts to align with the deployment's reasoning-heavy query workload.

ID	Evidence
Q85	“Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting.” (p.9)